

Latent Diffusion Model to Remove Motion Artifacts in Neonatal MRI

Quanhao Sun, Yaqiong Chai, Steven Li, Duan Xu, Michael Khoo, Hosung Kim

University of Southern California, Los Angeles, CA; University of California San Francisco, San Francisco, CA

Biomedical Engineering Society (BMES) Annual Meeting, 2025. Track: Biomedical Imaging and Instrumentation - Deep Learning and Artificial Intelligence in Biomedical Imaging and Instrumentation.

Introduction

Neonatal brain MRI is a critical imaging method for assessing early brain development and neurodevelopmental abnormalities and guiding clinical interventions. However, acquiring high-quality images in neonates is challenging because infants cannot remain still during image acquisition, and sedation is often not feasible. Their scans are thus frequently affected by motion artifacts. Neurodevelopmental features such as growing brains, incomplete cortical folding, ongoing myelination, and indistinct gray/white matter boundaries make the appearance of neonatal brains different from adult brains. Motion artifacts can easily obscure these important features and significantly impact downstream analyses.

Conventional motion correction approaches, including k-space interpolation, rigid-body registration, and convolutional neural network (CNN)-based post-processing, often perform poorly on neonatal data. Most approaches utilize simulated motions that are often too simplified or rely on paired data where motion-corrupted and clean images are available, which are difficult to obtain in pediatric imaging. These methods often lack robust modelling of various patterns of motion common in neonates.

Recently, generative diffusion models have emerged as a powerful alternative. They offer a probabilistic framework capable of learning data-driven priors and reconstructing clean images from noisy observations. In this study, we extend our previously developed Motion Adaptive Diffusion Model (MADM) to neonatal brain MRI. Trained on high-quality, artifact-free infant images, MADM learns the structural patterns of neonatal anatomy and predicts clean images from motion-corrupted images without needing paired supervision. In this study, we apply MADM to a very challenging task of head-motion correction on neonatal brain MRI and compare its performance with a CNN approach.

Materials and Methods

This study utilized a neonatal brain MRI dataset comprising 325 T1-weighted scans from 198 subjects, with gestational age at birth ranging from 26.7 to 44.3 weeks. All scans were acquired at UCSF Benioff Children's Hospital, San Francisco. Each scan was visually assessed for motion artifacts; 196 scans were labeled as clean and 128 as motion corrupted. Clean scans were used for model training, and motion-corrupted scans served as the testing set.

All images underwent intensity normalization to standardize voxel-wise signal distributions. The model operated on full 3D volumes rather than 2D slices to better capture spatially coherent motion patterns distributed across slices in neonatal anatomy.

We adopted the MADM, a latent generative framework previously validated in adult MRI motion correction. The architecture includes a Vector Quantized Generative Adversarial Network (VQGAN) for encoding and decoding, and a 3D latent diffusion model with an attention-enhanced U-Net for denoising. During training,

Gaussian noise was progressively added to the latent representations of clean images in the forward process. The reverse process removed noise conditioned on anatomical priors, reconstructing artifact-free images.

To evaluate the model's performance in the absence of ground truth (i.e., paired images with/without motion), we used NIQE (naturalness image quality evaluator) to assess perceptual quality and PSNR (peak signal-to-noise ratio) between input and output volumes to quantify denoising performance. All metrics were computed in 3D to capture whole-brain image quality. We compared the evaluation metrics between MADM and a previously developed CNN model.

Results, Conclusions, and Discussions

Our model demonstrated consistent improvements in image quality across two evaluation metrics. Compared to the original motion-corrupted inputs, the corrected images exhibited lower NIQE scores (reduced from 6.097 to 4.831 by 1.266 on average), indicating improved perceptual quality. PSNR increased by 2.806 dB, reflecting the model's ability to restore image intensity structure relative to the noisy input. These improvements were observed consistently across a range of motion severity and gestational age. We further compared our MADM with a baseline 3D convolutional neural network trained with simulated motion-corrupted images as input, corresponding images as ground truth. MADM consistently outperformed the CNN across all metrics. Specifically, MADM achieved lower NIQE (4.831 vs. 5.263) and higher PSNR (26.934 vs. 25.725 dB). Qualitatively, CNNs were less successful in recovering anatomical details in their output images and left residual artifacts in regions affected by severe motion. In contrast, MADM preserved finer details and sharper anatomical boundaries, particularly in key neurodevelopmental features such as the cortical plate, subplate, and early sulcal structures. These enhancements were visually evident in both mild and moderate motion cases, supporting the model's robustness to varying artifact severity.

The results demonstrate the capacity of our latent diffusion-based framework to perform effective motion artifact correction in neonatal brain MRI, even without access to paired clean references. By operating in a learned latent space conditioned on anatomical priors, the model preserves fine structural details while suppressing motion-induced degradation. The use of full 3D volumes further enables spatially coherent restoration, where motion artifacts often span multiple slices.

Clinically, this approach has the potential to avoid repetition of scans, improve diagnostic confidence, and enable more reliable downstream analysis to assess early brain development. Future work will focus on expanding the training dataset to include more varied motion patterns, incorporating simulated motion into the forward process, and evaluating model performance in the segmentation of neuroanatomy and the prediction of clinical outcomes. Overall, our findings support the use of diffusion-based generative models as a practical and robust solution to head-motion correction on neonatal brain MRI.

Acknowledgements. This study was supported by a USC Generative AI grant and a USC BME Mann Pilot grant (PR0002274).